RV College of Engineering

Experiential Learning

Synopsis

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Interlock: Geometric Part Identity and Contract-Validated Change Control

Team Details

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1. Introduction

Interlock is a database system for mechanical assembly data in which the identity of a part is computed from its geometry, the relationships between parts are declared as machine-checkable contracts, and every proposed change is validated against those contracts before it is admitted to the shared store. It is deliberately not a modeller: it ingests assemblies from existing CAD tools, identifies them, relates them, validates changes against them and answers structural questions about them.

It applies classical data structures to engineering data: a content-addressed hash table keyed by a geometric fingerprint for deduplication, a Merkle tree over the assembly for constant-time configuration comparison, a directed acyclic graph for structure, and recursive traversals and range-predicate joins for queries. The project addresses **SDG 9 (Industry, Innovation and Infrastructure)** by reducing integration rework, duplicate parts and late-stage redesign.

2. Problem Statement

When several teams build one machine, their parts meet for the first time at integration. A mounting hole moves two millimetres and nobody downstream is told; two parts are each within specification but their combined error is not; the same bracket exists under six part numbers. CAD tools represent shape but not ownership or dependency, while PLM tools track part numbers but treat geometry as an opaque attachment. The gap between them, where integration failures live, is filled today by spreadsheets and manual review.

Table 2.1: Key failure modes and where they are detected

Failure mode	Conventional detection	Interlock detection
Same shape stored under several part numbers	Manual catalogue review, often never	Fingerprint match at ingestion
Interface change not propagated	E-mail, or the integration build	Contract comparison at commit
Cross-team tolerance chain violated	Physical assembly	Chain re-evaluation at commit
Re-save reported as a change	File-byte comparison	Fingerprint unchanged; no-op
Concurrent edits to dependent parts	Locking, or last-write-wins	Merkle changed-path intersection

3. Objectives

- **Geometric identity:** a pose-invariant fingerprint (volume, area, sorted principal moments, surface-type histogram, topology counts), scaled and quantised so that a re-exported or re-posed solid resolves to one shape record.
- **Assembly structure:** a DAG over part, revision, shape and occurrence with a Merkle root hash and an append-only commit log, so configurations compare in constant time and history is recoverable.
- **Contracts and commit validation:** published interfaces (envelope, hole patterns, mass, CG window, datums) and an all-or-nothing commit pipeline that rejects a change while naming the violated constraint and affected team.
- **Tolerances and queries:** worst-case and RSS evaluation of declared tolerance chains, and explosion, where-used, impact and substitution queries answered without opening any solid.

4. Methodology

Six stages operate over one store with one strict rule: **the geometry kernel runs only at ingestion, never during a query.** Every geometric fact is extracted once and stored as indexed columns.

- **Ingestion:** a headless Open CASCADE kernel reads STEP (ISO 10303-242) files, yielding the occurrence tree, solids and transforms; cylinders (holes) and planes are recorded as candidate interfaces.
- **Geometric identity:** the invariant vector is scaled by part size, rounded and hashed; the digest is the primary key of the shape table, so duplicates collapse on write.
- **Assembly hashing:** each node hashes its sorted children with their transforms; a change rehashes only its path to the root, and differences are located by hash descent.
- **Contract matching:** a socket (what a parent needs) is matched to an interface (what a part offers) by pattern, fit class, envelope and budget checks, expressed as relational range joins.

- **Commit validation:** each change is classified as a no-op, internal or breaking change; concurrency is optimistic, using compare-and-swap on the root hash, with conflicts only when changed paths intersect.
- **Tolerances and queries:** chains are evaluated worst-case and RSS, cross-team chains are ranked highest, and all queries are recursive CTEs over indexed tables.

5. Comparison with Existing Systems

Table 5.1: Positioning against established tools

Tool category	Unit of identity	Point of enforcement
PDM / file vaults (SolidWorks PDM, Vault)	File and assigned part number	Manual review at check-in
PLM suites (Teamcenter, Windchill)	Human-assigned part number	Workflow approval, change orders
Git with large-file storage	Hash of file bytes	None; content is opaque
Shape search (Geolus, PARTsolutions)	Shape descriptor, for ranking only	None; results are advisory
Interlock (proposed)	Hash of a pose-invariant geometric vector	Automatic check inside the write path

Each mechanism exists somewhere today; the contribution is their composition: the shape descriptor is the primary key, the assembly hash is built from those keys, and contracts and tolerances are enforced at commit time rather than reported afterwards.

6. Expected Outcome

- Part identity established by geometry: importing two independent exports of one assembly creates no duplicate shape rows.
- Storage that grows with distinct geometry, not revision count, and structural queries answered in milliseconds without the solid modeller.
- False alarms suppressed: re-saves are no-ops, internal changes land silently, and only contract changes notify dependent teams.
- Integration errors caught at proposal: a commit that moves a mounting hole is rejected, naming the interface, chain and teams affected.
- Lock-free concurrent work, with the full build configuration identified by a single root hash.

7. Key References

- Osada, R., Funkhouser, T., Chazelle, B., and Dobkin, D. "Shape Distributions." ACM Trans. on Graphics, 21(4), 2002.
- Merkle, R. C. "A Digital Signature Based on a Conventional Encryption Function." CRYPTO '87, LNCS 293, Springer, 1988.
- ISO 10303-242:2020, Managed model-based 3D engineering (STEP AP242), ISO, Geneva, 2020.
- Jarratt, T. A. W., et al. "Engineering change: an overview and perspective on the literature." Research in Eng. Design, 22(2), 2011.
- Chase, K. W., and Parkinson, A. R. "A survey of research in the application of tolerance analysis to the design of mechanical assemblies." Research in Eng. Design, 3(1), 1991.